

Total No. of Questions : 8]

SEAT No. :

PB-2365

[Total No. of Pages : 2

[6263]-215

B.E. (Mechanical Engineering)
MODERN MACHINING PROCESSES
(2019 Pattern) (Semester - VII) (Elective - III) (402044C)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6 or Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Explain in detail with suitable sketch of electro chemical grinding. Also write merits, demerits and applications. **[9]**

b) Explain the chemical machining process with principle, construction, working process parameters and applications. **[8]**

OR

Q2) a) Explain in detail photochemical machining process with principle, construction, working process parameters. **[8]**

b) Differentiate electro chemical machining and photochemical machining process. **[5]**

c) Write in brief about electrolytes used in ECM process. **[4]**

Q3) a) Explain in detail with suitable sketch of electro chemical discharge machining. Also discuss the effect of process parameters, merits, demerits and applications. **[9]**

b) Explain with the suitable sketch principle used in EDM. Also describe in brief the effect of any two process parameter on material removal rate. **[9]**

OR

P.T.O.

- Q4)** a) Explain electric discharge diamond grinding with suitable sketch. [8]
b) Explain the principle of w- EDM with suitable sketch. [6]
c) Differentiate electrical discharge grinding and electro discharge diamond grinding. [4]

- Q5)** a) Explain micro turning with suitable sketch. Also discuss the advantages, disadvantages and applications. [8]
b) Explain micro drilling machining process with principle, construction, working process parameters and applications. [9]

OR

- Q6)** a) Explain with suitable sketch diamond turn machining process. [8]
b) Explain micro EDM tool manufacturing with suitable sketch. [5]
c) Compare micro milling and micro drilling. [4]

- Q7)** a) Explain Magnetic Abrasives Finishing with principle, construction, working process parameters and applications. [9]
b) Explain Magnetorheological Finishing with principle, construction, working process parameters and applications. [9]

OR

- Q8)** a) Explain photolithography process in details. [8]
b) Explain microsystem with suitable application. [6]
c) Compare Magnetorheological Finishing and Magnetic Abrasives Finishing. [4]

